SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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CSA37- SOFTWARE TESTING

EXP:4

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Requirement Analysis and Positive Test Scenarios and Test Cases for an E-Commerce Application

Aim

To study the requirements of an E-Commerce Application, and to prepare positive test scenarios and test cases that confirm the application works correctly for valid inputs and normal user actions.

Problem Statement

An e-commerce application lets customers browse products, add them to a cart, pay online and track their orders. It handles personal data, payments and stock, so a defect can cause lost sales, wrong charges or loss of customer trust.

Testing is needed to check that the application:

- Customers can register, log in, search for products and place orders correctly.
- Invalid actions are rejected safely with clear messages.
- Cart, payment and order handling follow the stated requirements.

Any behaviour that differs from the specification must be recorded, with enough detail for developers and testers to repeat and fix it.

Software Requirements

1 Functional requirements (system specification)

ID	Requirement
FR1	A user can register with name, email, mobile number and password. The email must be unique.
FR2	A registered user can log in with valid credentials and log out.
FR3	The user can search products by name or category and get relevant results.
FR4	The user can filter and sort results by price, brand and rating.
FR5	The product page shows images, price, description, rating and stock availability.
FR6	The user can add a product to the cart, change its quantity and remove it. Out-of-stock items cannot be added.
FR7	The cart shows the correct item prices and total. A valid coupon reduces the total correctly.
FR8	Checkout needs a valid delivery address. The user can save more than one address.
FR9	Payment is possible by card, UPI, net banking or cash on delivery.
FR10	After a successful payment, the system shows an order ID and sends an email or SMS confirmation.
FR11	The user can track the order status and view the order history.
FR12	An order can be cancelled before it is shipped. The refund goes to the original payment method.

2 Software and environment requirements

Item	Details
Application	Web application (React front end, Java/Node.js back end) and mobile browser view
Database	MySQL / MongoDB (users, products, carts, orders)
Operating system	Windows 10/11 or Linux
Browsers	Chrome, Firefox, Edge (latest versions)
Testing tools	Test case document (Excel), Selenium, Postman, bug tracker (Jira)
Test data	Sample users, products with and without stock, coupon codes, sandbox card and UPI details

Positive Test Scenarios

A positive test scenario checks that a feature works as expected when the user gives valid input and follows the normal flow.

Scenario ID	Module	Test Scenario	Requirement
EC-PS01	Registration	Register with valid details	FR1
EC-PS02	Login	Log in with valid credentials	FR2
EC-PS03	Search	Search for a product by name	FR3
EC-PS04	Filter and sort	Filter by price and sort low to high	FR4
EC-PS05	Product page	View product details	FR5
EC-PS06	Cart	Add a product to the cart	FR6
EC-PS07	Cart	Change the quantity and remove an item	FR6
EC-PS08	Cart	Apply a valid coupon	FR7
EC-PS09	Checkout	Proceed with a saved address	FR8
EC-PS10	Payment	Pay using UPI	FR9
EC-PS11	Payment	Place an order with cash on delivery	FR9, FR10
EC-PS12	Order tracking	Track an order and view history	FR11
EC-PS13	Cancellation	Cancel an order before shipping	FR12

Test Cases

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
EC-PT01	EC-PS01	1. Open the registration page 2. Enter name, unique email, mobile number and password 3. Click Register	Ravi Kumar; ravi.k@example.com; 9876543210; Ravi@1234	Account is created, a success message is shown and a verification email is sent.
EC-PT02	EC-PS02	1. Open the login page 2. Enter the registered email and password 3. Click Login	ravi.k@example.com; Ravi@1234	User is logged in and the home page shows the user's name.
EC-PT03	EC-PS03	1. Type the product name in the search box 2. Press Enter	Wireless mouse	Relevant products are listed with image, name and price.
EC-PT04	EC-PS04	1. Search for a product 2. Set the price range and brand filter 3. Select Sort by Price: Low to High	Headphones; ₹500 to ₹2,000	Only matching products are shown, in ascending order of price.
EC-PT05	EC-PS05	1. Click a product from the search results	Any in-stock product	Images, price, description, rating and stock status are displayed with an Add to Cart button.
EC-PT06	EC-PS06	1. Open an in-stock product 2. Click Add to Cart 3. Open the cart	Wireless mouse; quantity 1	Item is added, the cart count becomes 1 and the price is correct.
EC-PT07	EC-PS07	1. Open the cart 2. Change the quantity to 3 3. Check the total 4. Click Remove	Quantity 3	Total is recalculated as 3 × price. After removal the cart shows an empty message.
EC-PT08	EC-PS08	1. Add items worth ₹1,500 2. Enter the coupon in the cart 3. Click Apply	Coupon SAVE10 (10% off)	A discount of ₹150 is applied and the new total is ₹1,350.
EC-PT09	EC-PS09	1. Click Proceed to Checkout 2. Select a saved address 3. Click Continue	Saved home address	Address is accepted and the order summary shows the delivery date.
EC-PT10	EC-PS10	1. Select UPI 2. Enter the test UPI ID 3. Approve the payment in the sandbox	success@upi	Payment succeeds and the order is placed.

Test Case ID	Scenario ID	Test Steps	Test Data	Expected Result
EC-PT11	EC-PS11	1. Select Cash on Delivery 2. Click Place Order 3. Check the confirmation page and email	Order value ₹799	Order ID is shown, and an email or SMS confirmation is received.
EC-PT12	EC-PS12	1. Open My Orders 2. Select the latest order 3. Click Track Order	Order placed in the previous test	Current status and expected delivery date are shown, and the order appears in the history.
EC-PT13	EC-PS13	1. Open My Orders 2. Select an order with status Confirmed 3. Click Cancel and choose a reason 4. Confirm	Prepaid order	Order is cancelled and a refund to the original payment method is initiated.

Result / Conclusion

Result: 13 positive test scenarios and 13 test cases were prepared for the E-Commerce Application, covering 12 of the 12 functional requirements (FR1 to FR12). The test cases by module are:

Module	Test Scenarios	Test Cases
Registration	1	1
Login	1	1
Search	1	1
Filter and sort	1	1
Product page	1	1
Cart	3	3
Checkout	1	1
Payment	2	2
Order tracking	1	1
Cancellation	1	1
Total	13	13

Conclusion: The positive test cases cover the main user flows of the E-Commerce Application, from the first action to the final result. When they are executed on a build, they confirm that the

valid-input behaviour matches the specification. They should be run first in every test cycle and kept in the regression suite, with the negative test cases added to check invalid inputs.